# MARLENE D. BERKE

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## EDUCATION AND APPOINTMENTS

**MIT** *2025 - present*
Postdoctoral Fellow, Integrative Computational Neuroscience Center
Advisors: Rebecca Saxe & Joshua Tenenbaum

**Yale University** *2019 - 2025*
PhD in Cognitive Psychology
Advisor: Julian Jara-Ettinger
Dissertation: Mental Representations of Computational Processes

**Cornell University** *2015 - 2019*
BA in Computational Cognitive Neuroscience (College Scholar)
Minors in Statistics, Biology, Psychology, and Cognitive Science
GPA: 4.0.

## JOURNAL ARTICLES (PUBLISHED)

**Berke, M.**, & Casser, L. (*in press*). Bodily Core Knowledge. Commentary on Bai et al. *Behavioral and Brain Sciences.*

Krapf, J., Yong, P., **Berke, M.**, Bohm-Starke, N., Bornstein, J., Chrysilla, E., Dempsey, T., Falsetta, M., Foster, D., Goldstein, S., Iadarola, M., Kellogg-Spadt, S., Mannes, A., Vogel, J., & Goldstein, A. (2025). Key Outcomes of the Vulvodynia Therapeutic Research Summit: Identifying Targets for Further Research Study in the Treatment of Provoked Vestibulodynia. *Obstetrics & Gynecology.* link

Rubio-Fernandez, P., **Berke, M.**, & Jara-Ettinger, J. (2025). Tracking minds in communication. *Trends in Cognitive Sciences.* link

**Berke, M.**, & Jara-Ettinger, J. (2024). Core knowledge, visual illusions, and the discovery of the self. Commentary on Spelke. *Behavioral and Brain Sciences.* link

**Berke, M.**, Walter-Terrill, R., Jara-Ettinger, J., & Scholl, B. J. (2022). Flexible goals require that inflexible perceptual systems produce veridical representations: Implications for realism as revealed by evolutionary simulations. *Cognitive Science.* link

## JOURNAL ARTICLES (UNDER REVIEW / IN PREPARATION)

**Berke*, M.**, Horschler*, D.J., Royka, A., Santos, L.R., & Jara-Ettinger, J. (R&R, *Cognition*). Differences between human and non-human primate Theory of Mind: Evidence from computational modeling. preprint

**Berke, M.**, Tenenbaum, A., Sterling, B., & Jara-Ettinger, J. (R&R, *Psychological Review*). People reconstruct others' cognitive processes by representing their minds as computational systems.

**Berke, M.**, Sterling, B., Tan, Z., Chandra, K., & Jara-Ettinger, J. (*under review*). People craft lies to exploit their audience's beliefs, desires, and reasoning errors.

**Berke, M.**, Collins, K., Tenenbaum, J., & Saxe, R. (*in prep*). Toward a formalization of human intuitive theories of bodily pain.

## CONFERENCE PROCEEDINGS

**Berke, M.**, Collins, K., Tenenbaum, J., & Saxe, R. (2026). Toward a formalization of human intuitive theories of bodily pain. *Proceedings of the 48th Annual Conference of the Cognitive Science Society.* preprint

Royka, A., Townrow, L., Baker, C., **Berke, M.**, & Santos, L. (2026). Monkeys fail inference versions of classic intuitive physics prediction tasks. *Proceedings of the 48th Annual Conference of the Cognitive Science Society.*

Collins, K., **Berke, M.**, Sucholutsky, I., Ali, A., Weller, A., O'Donnell, T., Brooke-Wilson, T.*, Wong, L.*, & Tenenbaum, J.* (2026). Medical model synthesis architectures: A case study. *AI4GOOD Workshop @ICML 2026.* (*equal senior advising) preprint

**Berke, M.***, Sterling, B.*, Chandra, K., & Jara-Ettinger, J. (2025). People use theory of mind to craft lies exploiting audience desires. *Proceedings of the 47th Annual Conference of the Cognitive Science Society.* link

Zhang, R., **Berke, M.**, & Jara-Ettinger, J. (2025). Six-year-olds use an intuitive theory of attention to infer what others see, whom to trust, and what they want. *Proceedings of the 47th Annual Conference of the Cognitive Science Society.* link

Shachnai, R., Kleiman-Weiner, M., **Berke, M.**, & Leonard, J. (2025). When Bayesians take over: A computational model of parental intervention. *Proceedings of the 47th Annual Conference of the Cognitive Science Society.* link

**Berke, M.**, Azerbayev, Z., Belledonne, M., Tavares, Z., & Jara-Ettinger, J. (2024). MetaCOG: A hierarchical probabilistic model for learning meta-cognitive visual representations. *Proceedings of the 40th Conference on Uncertainty in Artificial Intelligence.* link

**Berke, M.**, Sterling, B., Tenenbaum, A., & Jara-Ettinger, J. (2024). No signatures of first-person simulation in Theory of Mind judgments about thinking. *Proceedings of the 46th Annual Conference of the Cognitive Science Society.* link

Tan, Z. T., Jara-Ettinger, J., & **Berke, M.** (2024). Reasoning about knowledge in lie production. *Proceedings of the 46th Annual Conference of the Cognitive Science Society.* link

**Berke, M.**, Tenenbaum, A., Sterling, B., & Jara-Ettinger, J. (2023). Thinking about thinking as rational computation. *Proceedings of the 45th Annual Conference of the Cognitive Science Society.* link

**Berke, M.**, & Jara-Ettinger, J. (2022). Integrating experience into Bayesian Theory of Mind. *Proceedings of the 44th Annual Conference of the Cognitive Science Society.* link

**Berke, M.**, & Jara-Ettinger, J. (2021). Thinking about thinking through inverse reasoning. *Proceedings of the 43rd Annual Conference of the Cognitive Science Society.* link

**Berke, M.**, Field, D., & Cleland, T. (2017). The sparse structure of natural chemical environments. *Proceedings from the International Symposium on Olfaction and Electronic Nose (ISOEN).* link

## IN POPULAR PRESS

Hintz, E. & **Berke, M.** (2025). When doctors don't believe their patients' pain — experts explain the all-too-common experience of medical gaslighting. *The Conversation.* Syndicated in CNN and other outlets. link

## CONFERENCE PRESENTATIONS

**Berke, M.**, Collins, K., Tenenbaum, J., & Saxe, R. (2026). Toward a formalization of human intuitive theories of bodily pain. Talk at the 52nd annual meeting of the Society for Philosophy and Psychology. Poster also presented at the 48th Annual Conference of the Cognitive Science Society.

**Berke, M.***, Sterling, B.*, Chandra, K., & Jara-Ettinger, J. (2025). People use theory of mind to craft lies exploiting audience desires. Talk at the 51st annual meeting of the Society for Philosophy and Psychology. Talk also presented at the 47th Annual Conference of the Cognitive Science Society.

**Berke, M.***, Horschler, D.J.*, Royka, A., Santos, L.R., & Jara-Ettinger, J. (2025). A new computational approach to a classic question: How human-like is non-human primate theory of mind? Talk at the 51st annual meeting of the Society for Philosophy and Psychology.

**Berke, M.**, Azerbayev, Z., Belledonne, M., Tavares, Z., & Jara-Ettinger, J. (2024). MetaCOG: A hierarchical probabilistic model for learning meta-cognitive visual representations. Poster at the 40th Conference on Uncertainty in Artificial Intelligence.

Tan, Z. T., Jara-Ettinger, J., & **Berke, M.** (2024). Reasoning about Knowledge in Lie Production. Talk at the 50th annual meeting of the Society for Philosophy and Psychology. Talk also presented at the 46th Annual Conference of the Cognitive Science Society.

**Berke, M.**, Sterling, B., Tenenbaum, A., & Jara-Ettinger, J. (2024). No signatures of first-person biases in Theory of Mind judgments about thinking. Talk at the 46th Annual Conference of the Cognitive Science Society.

**Berke, M.**, Tenenbaum, A., Sterling, B., & Jara-Ettinger, J. (2023). Thinking about thinking as rational computation. Poster at the 49th annual meeting of the Society for Philosophy and Psychology. Poster also presented at the 45th Annual Conference of the Cognitive Science Society.

**Berke, M.**, & Jara-Ettinger, J. (2022). Integrating experience into Bayesian Theory of Mind. Poster presented at the 44th Annual Conference of the Cognitive Science Society.

**Berke, M.**, Azerbayev, Z., Belledonne, M., Tavares, Z., & Jara-Ettinger, J. (2022). MetaCOG: Learning a metacognition for what objects are actually there. Beyond Bayes: Paths Towards Universal Reasoning Systems Workshop @ICML 2022.

**Berke, M.**, Azerbayev, Z., Belledonne, M., Tavares, Z., & Jara-Ettinger, J. (2022). MetaCOG: Learning a metacognition for what objects are actually there. Poster at the Neurosymbolic Programming summer school at Caltech.

**Berke, M.**, & Jara-Ettinger, J. (2021). Thinking about thinking through inverse reasoning. Poster at the 43rd Annual Conference of the Cognitive Science Society and the 47th Annual Meeting of the Society for Philosophy and Psychology.

**Berke, M.**, Walter-Terrill, R., Jara-Ettinger, J., & Scholl, B. (2021). Flexible goals require that inflexible perceptual systems produce veridical representations: Implications for realism as revealed by evolutionary simulations. Talk at the 21st Annual Meeting of the Vision Science Society.

**Berke, M.**, Belledonne, M., & Jara-Ettinger, J. (2020). Learning a metacognition for object perception. Poster at Shared Visual Representations in Human & Machine Intelligence (SVRHM) @NeurIPS 2020.

## SELECTED INVITED TALKS

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| Philosophy of Medicine Workshop, Dartmouth College | *Fall 2025* |
| Thought Dynamics Lab, Hebrew University of Jerusalem, Israel | *Spring 2025* |
| NYUConcats, New York University | *Spring 2025* |
| Saxelab, Massachusetts Institute of Technology | *Fall 2024* |
| Causality in Cognition Lab, Stanford University | *Fall 2024* |
| Computational Cognitive Science Group, Massachusetts Institute of Technology | *Spring 2023* |
| Human and Machine Cognition Lab, Max Planck Institute for Biological Cybernetics Tübingen, Germany. | *Spring 2022* |

## FELLOWSHIPS AND ACADEMIC HONORS

**Integrative Computational Neuroscience Postdoctoral Fellowship** *2025 - 2028*
Proposed "Computational Cognitive Models of Pain Communication" project. Competitively awarded three-year postdoctoral fellowship.

**Graduate Student Assembly Conference Travel Fellowship** *Summer 2022*
To attend *International Conference on Machine Learning* (ICML)

**Sterling Prize** *Fall 2019 - Spring 2021*
Awarded the prestigious Sterling Prize, totaling $7000

**College Scholar Honors Thesis** *Fall 2018 - Spring 2019*
Awarded Summa Cum Laude

**Tanner Dean Scholar** *2015 - 2019*
Granted a $5000 research stipend for Summer 2018

## TEACHING

**Computational Cognitive Science** *Fall 2025*
Teaching Assistant (MIT). Supervised undergraduate and master's students' final projects.

**Assistant Director of Undergraduate Studies** *Fall 2024 - Spring 2025*
Assisted in administering the cognitive science and psychology majors and advised students on individualized plans of study in cognitive science.

**Algorithms of the Mind** *Fall 2023*
Delivered guest lecture "Social Cognition as Inverse Planning" on modeling Theory of Mind as Bayesian inferences over POMDPs.

**Introduction to Statistics (2x)** *Spring 2021 & 2022*
Teaching Fellow (Yale University). Led discussion sections. Avg. student rating: 4.3/5

**Multivariate Statistics (2x)** *Fall 2020 & 2021*
Teaching Fellow (Yale University). Delivered guest lectures on advanced computational methods. Avg. student rating: 4.75/5

**Introduction to Statistics for Biology (4x)** *Fall 2017 - Spring 2019*
Teaching Assistant (Cornell University). Led discussion sections.

## MENTORSHIP

**Yale Psychology Sneak Peek Program Mentor** *Fall 2020 - Fall 2024*
The program pairs individuals from an underrepresented background with a current PhD student to help them apply to graduate school.

**NextGen Psych Scholars Program Mentor** *Summer 2023 - Spring 2025*
This inter-university program pairs undergraduates and post-bacs from underrepresented backgrounds with a current PhD student to help them apply to graduate school.

**Yale Psychology Diversity Committee Mentor** *Fall 2020 - Fall 2024*
The program pairs incoming PhD students with current PhD student mentors.

**Undergraduates Research Assistants Mentored**
Zhangir Azerbayev (December 2020 - May 2022; now at OpenAI), Tanushree Burman (January 2021 - Fall 2021; now PhD student in computer science at Tufts), Bernardo Eilert Trevisan (January 2021 - Spring 2021), and Tan Zhi Yi (May 2022 - May 2024; Honors Thesis student based in the Yale-NUS Singapore campus), Abi Tenenbaum (June 2022 - May 2023), Ben Sterling (June 2022 - May 2025; Honors Thesis in Cognitive Science; now PhD student in Psychology at NYU).

## SERVICE

**Reviews**
Ad-hoc reviewer for *Psychological Review*, *Cognition*, *Proceedings of the Royal Society B*, *Open Mind*, the 43rd–48th *Annual Conference of the Cognitive Science Society*, Social Intelligence in Humans and Robots workshop @RSS 2024, From Human Cognition to AI Reasoning workshop @ICLR 2026, Program committee member for AI meets Moral Philosophy and Moral Psychology @NeurIPS 2023.

**Yale Psychology Interview Weekend**
Organized interview weekend for Yale Psychology's cognitive area.

## ADVOCACY AND PUBLIC SERVICE

**Medical Advocacy**
Established the Connecticut Chapter of Tight Lipped, a medical advocacy organization by and for people with chronic pelvic, vulvar, and vaginal pain conditions. I collaborated with Yale Medicine and UCONN Health OBGYN departments to improve clinician education on these conditions. We also partnered with Yale Health to identify bottlenecks in care, ultimately leading to changes in processes impacting 60,000 patients on the Yale Health plan.

**Patient Representative at Vulvodynia Summit 2024**
Presented on the key barriers to care for chronic vulvar and vaginal pain. This was an invite-only summit of international vulvar pain researchers held in Washington, DC.

**Assessment of NIH Research on Women's Health**
Testified in the Stakeholder Perspectives session of National Academies of Sciences, Engineering, and Medicine's Assessment of NIH Research on Women's Health in January 2024, urging that chronic pelvic and vulvovaginal pain be identified as a high priority area for research. My testimony was included in the NASEM's report.

**Connecticut House Bill: An Act Promoting Equity In Coverage For Fertility Health Care**
I testified in the hearing on Connecticut House Bill 6617, *An Act Promoting Equity In Coverage For Fertility Health Care* in February 2023, which would have required health insurance companies in Connecticut to cover fertility treatments. I urged that chronic vulvovaginal pain conditions be recognized as impacting fertility, so that insurance companies be required to cover treatment for chronic vulvovaginal pain conditions.